Computational Biology

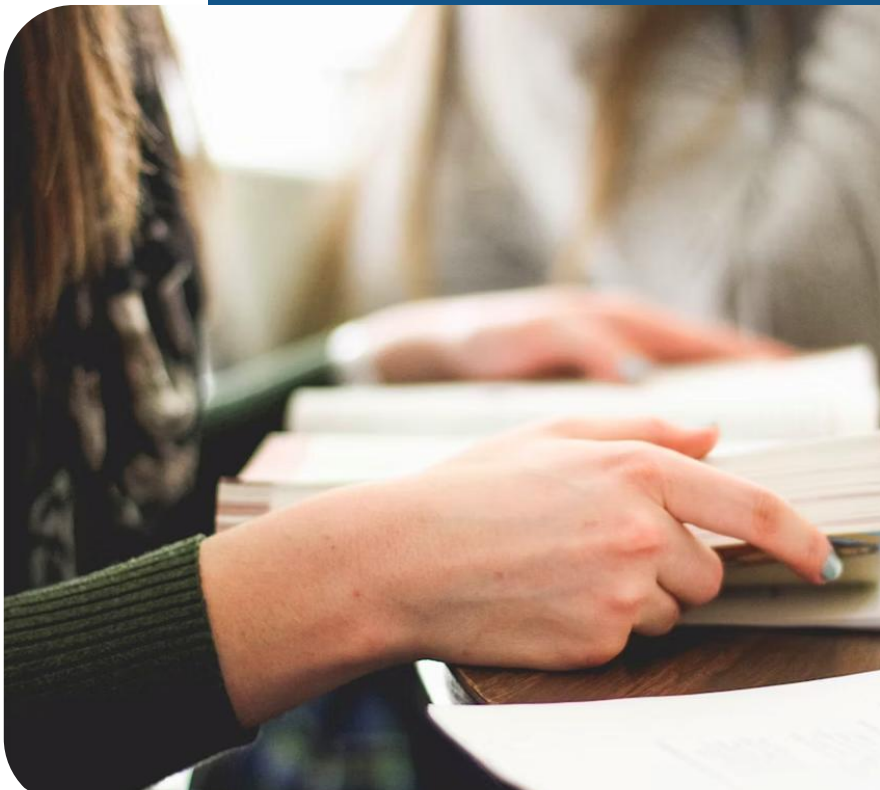
Session 3



Today's Agenda

- Evolution
- DNA Mutation
- DNA Sequencing
- PCR

Evolution



DIFFERENCES IN:



+



GENES ENVIRONMENT

EOHIPPUS



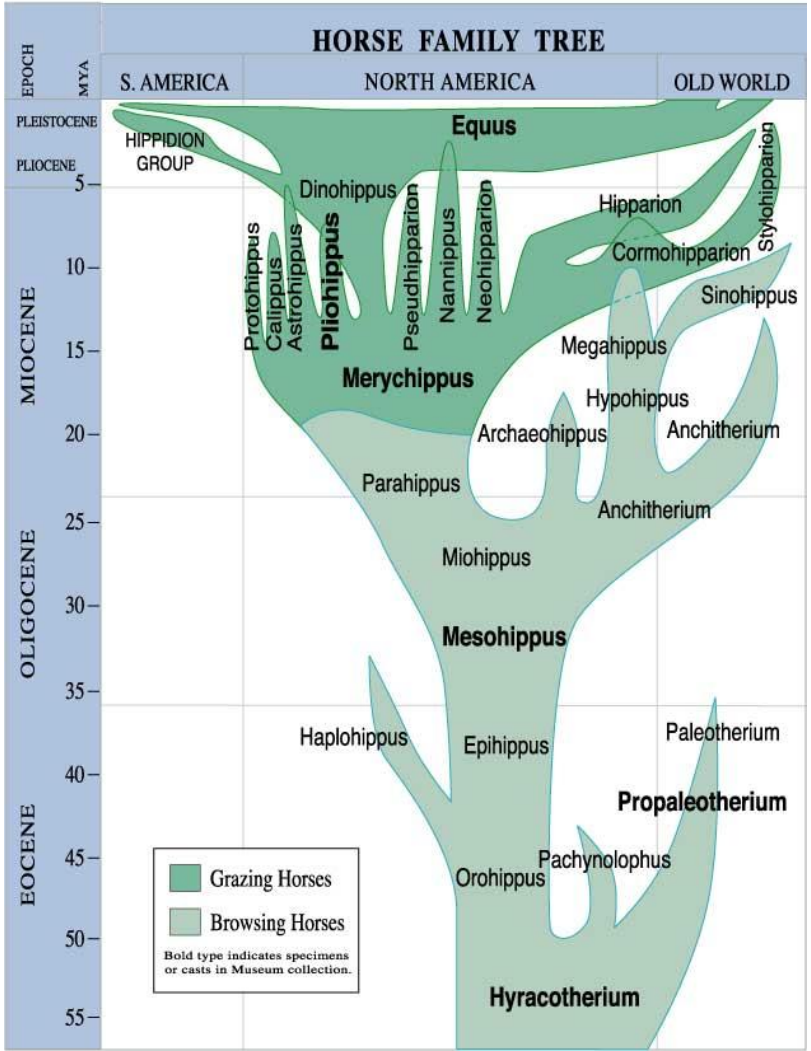
VARIATION BETWEEN INDIVIDUALS

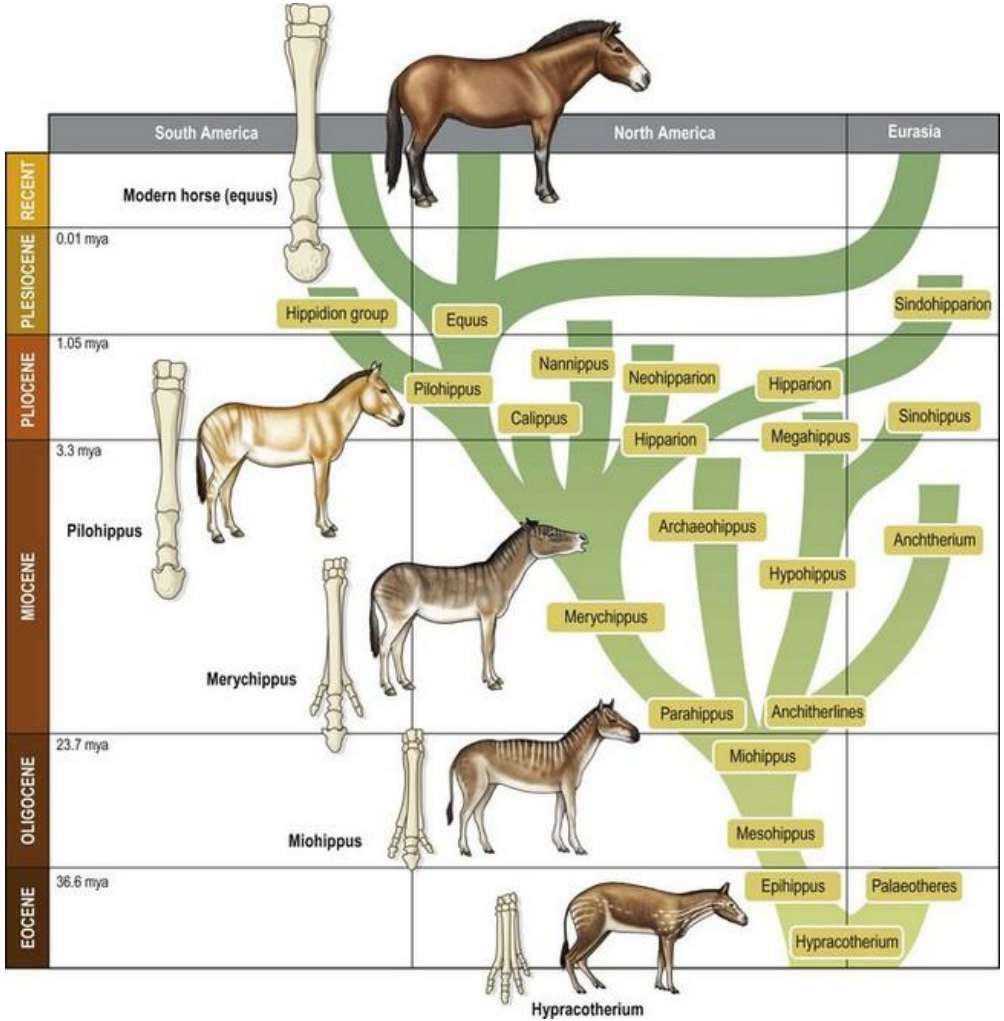


HORSE



EVOLUTION OF NEW SPECIES


HORSE FAMILY TREE
EPOCH
MYA
S. AMERICA
NORTH AMERICA
OLD WORLD
PLEISTOCENE
PLIOCENE
MIOCENE
OLIGOCENE
EOCENE
5
10
15
20
25
30
35
40
45
50
55
HIPPIDION GROUP
Equus
Dinohippus
Protohippus
Calippus
Astrohippus
Pliohippus
Pseudhipparion
Nannippus
Neohipparion
Hipparion
Cormohipparion
Stylohipparion
Sinohippus
Megahippus
Merychippus
Hypohippus
Archaeohippus
Anchitherium
Parahippus
Anchitherium
Miohippus
Mesohippus
Haplohippus
Epihippus
Paleotherium
Propaleotherium
Pachynolophus
Orohippus
Hyracotherium
Grazing Horses
Browsing Horses
Bold type indicates specimens or casts in Museum collection.


South America
North America
Eurasia
RECENT
PLESIOCENE
PLIOCENE
MIOCENE
OLIGOCENE
EOCENE
Modern horse (equus)
0.01 mya
1.05 mya
3.3 mya
23.7 mya
36.6 mya
Hippidion group
Equus
Sindohipparion
Nannippus
Neohipparion
Hipparion
Pilohippus
Calippus
Sinohippus
Megahippus
Archaeohippus
Anchtherium
Hypohippus
Merychippus
Parahippus
Anchitherlines
Miohippus
Mesohippus
Epihippus
Palaeotheres
Hypracotherium
Pilohippus
Merychippus
Miohippus
Hypracotherium



POPULATION

- ALL DIFFERENT (VARIATION)

DIFFERENT PHENOTYPES:

- OBSERVABLE CHARACTERISTICS



- LOTS OF DIFFERENT GENES



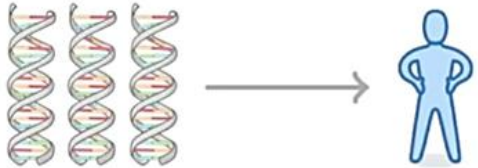
- LOTS OF DIFFERENT ENVIRONMENTAL FACTORS



GENES



- CODE FOR PROTEINS
- GENOME = ALL YOUR GENES COMBINED



(CODES FOR AN ENTIRE ORGANISM)



- EVERYONE HAS A UNIQUE GENOME (EXCEPT IDENTICAL TWINS)
- DIFFERENT PROTEINS, LOOK DIFFERENT

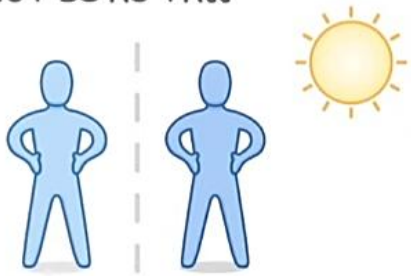
ENVIRONMENT



IDENTICAL TWINS, SAME GENES



- DIDN'T EAT OR SLEEP AS MUCH: MAY NOT BE AS TALL

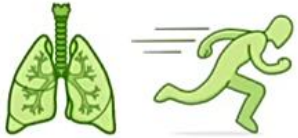


- MORE TIME IN THE SUN: DARKER SKIN

PROTEIN MAY BE DIFFERENT

- PHENOTYPE WILL CHANGE SLIGHTLY
- USUALLY SOMETHING BAD
- VERY OCCASIONALLY BENEFICIAL:
  - RESISTANT TO LUNG DISEASE
  - ABILITY TO RUN FASTER
- MORE LIKELY TO SURVIVE
- MORE LIKELY TO REPRODUCE (PASS ON GENES)

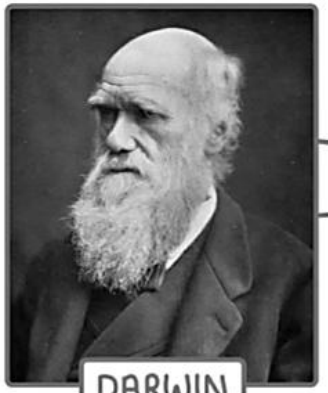






MUTATIONS

CHANGE IN THE DNA CODE



DARWIN

'SURVIVAL OF THE FITTEST'

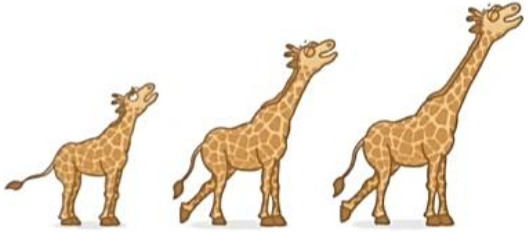
TRAITS PASSED FROM PARENT TO CHILD (USEFUL TRAITS PASSED ON THE MOST)

NATURAL SELECTION

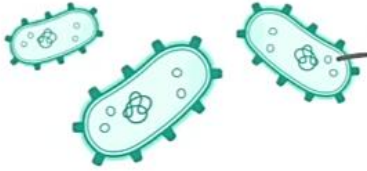
FITTEST INDIVIDUALS SELECTED TO SURVIVE

EVOLUTION

INHERITENCE OF CERTAIN CHARACTERISTICS IN A POPULATION, OVER MULTIPLE GENERATIONS, COULD LEAD TO A CHANGE IN THE WHOLE SPECIES



(OR DEVELOPMENT OF A NEW SPECIES)



FIRST DEVELOPED MORE THAN 3 BILLION YEARS AGO

ALL LIVING SPECIES EVOLVED FROM SIMPLE LIFE FORMS



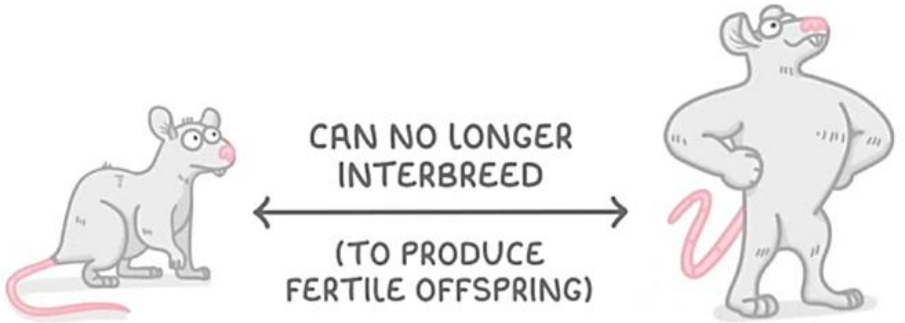


ALL LIVING SPECIES MUST HAVE EVOLVED FROM DIFFERENT SPECIES IN THE PAST

EVOLUTION

NATURAL SELECTION OF CERTAIN GENETIC CHANGES

GIVE RISE TO PHENOTYPES THAT ARE BEST SUITED FOR THE ENVIRONMENT



NEW SPECIES